Tanmay Bhuskute

+1 (732) 858-3027 | tanmay.v.bhuskute@gmail.com | github.com/Tanmayb05 | linkedin.com/in/tanmay-bhuskute
Cincinnati, OH

SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) working at the intersection of security engineering and AI/ML: production experience in secure infrastructure automation, IAM/secrets management, and CI/CD quality gates, paired with graduate research building and adversarially evaluating LLM-driven automation pipelines.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, TypeScript, C/C++, SQL, Go, Bash.

Security & Secure SDLC: SonarQube quality gates, HashiCorp Vault, IAM policy design, secure CI/CD, code coverage enforcement.

AI & ML: Gemini, Vertex AI, MCP (Model Context Protocol), PyTorch, TensorFlow, scikit-learn, LangChain, RAG pipelines.

DevOps & Cloud: Git, GitLab CI/CD, Jenkins, AWS, Docker, Kubernetes, ArgoCD, Helm, Terraform, Linux, Azure, Google Cloud Platform.

Web & Backend: React, Node.js, Django, Flask, FastAPI, REST APIs, PostgreSQL, MySQL.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, benchmarked against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Built provider-agnostic LLM abstraction spanning **Gemini, GPT-4o, Llama, and Qwen** to isolate method robustness from any single model's behavior, demonstrating **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Managed **HashiCorp Vault** secrets and IAM access policies across production and non-production environments, strengthening security posture for **40+** cloud environments.
- Served as Quality Coach and SonarQube Specialist, enforcing secure coding standards and CI/CD quality gates that increased automated code coverage from **35% to 78%**; mentored **5 engineers** on secure development practices.
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, GitLab CI/CD, and HashiCorp Vault, designing an Ansible-based Blue-Green framework that cut upgrade downtime by **50%**.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** validation framework, catching **95%** of misconfigurations pre-production.

PROJECTS AND RESEARCH

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Built output-handling control layer over Google Gemini 2.5-flash's **RAG pipeline**, forcing citation-backed, evidence-grounded answers to prevent ungrounded/hallucinated output reaching high-stakes healthcare eligibility decisions.
- Designed **17-tool MCP server** for external AI agent tool-calling with **Zod schema validation at every boundary**, rejecting malformed or injected tool inputs before execution.

NYC Mobility Forecasting [PySpark, Delta Lake, scikit-learn, FastAPI] Jul 2026

- Built ML system forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** hourly dataset); designed two-stage Poisson hurdle model improving MAPE **18.77% to 16.52%** and cutting RMSE **43%** over baseline Random Forest.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).